

Building Agents with OpenAI's AgentKit and Responses API

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The Problem

Building Production Agents Is Hard

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OpenAI's answer: a layered stack — each layer builds on the previous one.

The Three-Layer Stack



The Three-Layer Stack



Each layer builds on the previous — you pick the level of abstraction you need.

Layer 1: Responses API

The Primitive

Responses API — What & Why

Released **March 2025**. Replaces both Chat Completions API and Assistants API.

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Key Innovations

- **Items-based I/O** — polymorphic output (text, tool calls, reasoning)
- **Server-side state** via `previous_response_id`
- **Reasoning preservation** across turns

Built-in Tools

- `web_search`
- `file_search`
- `code_interpreter`
- `computer_use`
- `mcp` (Model Context Protocol)

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This is what we've been using in the notebooks so far.

^[1]Why we built the Responses API

Layer 2: Agents SDK

The Framework

Agents SDK — Overview

Open-source framework. **Python + TypeScript.**

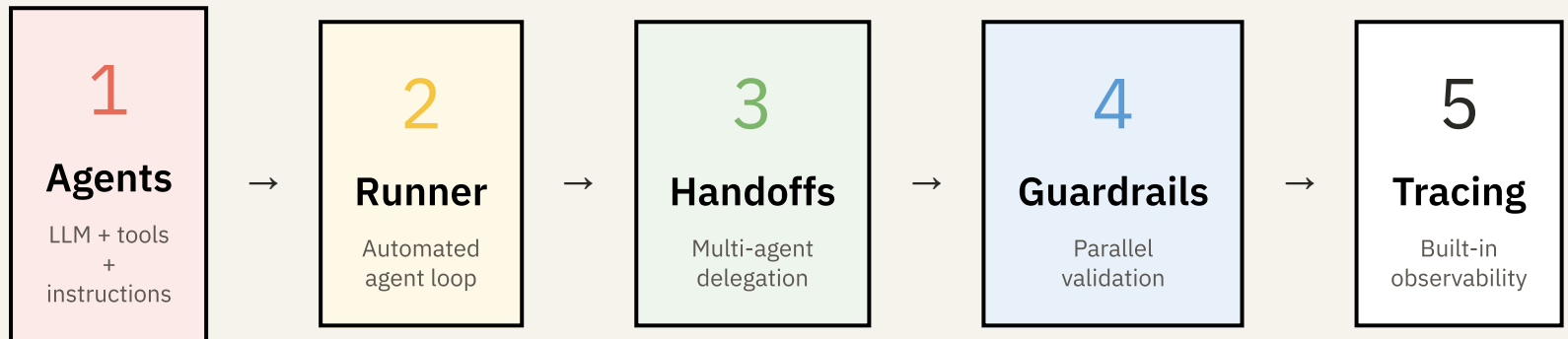
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pip install openai-agents
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The 5 Core Primitives

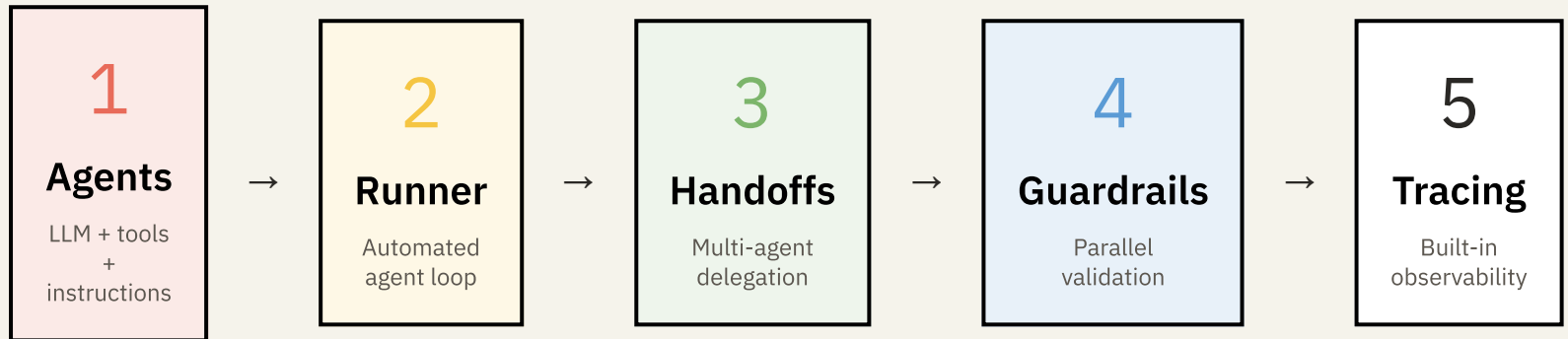


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The 5 Core Primitives



Works with **100+ LLMs** — not locked to OpenAI.

[1] [openai-agents-python \(GitHub\)](#).

[2] [Agents SDK Documentation](#)

Layer 3: AgentKit

The Product Platform

AgentKit — The Full Platform

Launched at **DevDay, October 2025**. No platform fee — standard API pricing.

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Agent Builder

Visual drag-and-drop canvas for composing multi-agent workflows



ChatKit

Embeddable chat UI — React components + Web Component



Connector Registry

Pre-built data connectors — Dropbox, Google Drive, SharePoint, MCP



Evals for Agents

Datasets, trace grading, automated prompt optimization

^[1][Introducing AgentKit — OpenAI](#)

Agent Builder

Visual Workflow Orchestration

Agent Builder — Deprecation Notice



Agent Builder & Evals are being shut down

OpenAI will retire **Agent Builder** and **Evals** on **November 30, 2026**.

The demos in this module will not be reproducible after that date.

The patterns you are learning — multi-agent orchestration, handoffs, guardrails, classifiers, and retrieval — are the **fundamental building blocks**. They are not tied to this UI. When the tooling changes, the thinking stays.

Source: community.openai.com/t/deprecation-notice-agent-builder/1382650

Where the patterns live after November 2026

Agents SDK

```
pip install openai-agents
```

- Same patterns in Python code
- Handoffs, guardrails, file search as classes
- Full developer control, self-hosted

Workspace Agents

ChatGPT Business / Enterprise

- No-code, connects to Slack / Google Drive / Salesforce
- Internal team automation — not a developer API

Module rewrite targeting Agents SDK is planned for a future course revision.

Agent Builder — What It Is

A **visual drag-and-drop canvas** for composing multi-step agent workflows — no orchestration glue code required.

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Part of AgentKit

- Sits alongside **ChatKit**, **Connector Registry**, and **Evals**
- Powered by the **Responses API** under the hood
- Standard API pricing — no platform fee

What You Get

- Build multi-agent workflows visually
- Test interactively before deploying
- Publish and version with one click
- Connect to ChatKit or export as code

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Target users: developers who want visual orchestration *and* teams who need to ship quickly without writing agent loop code.

Core Concepts — Workflows & Nodes

Workflow

The blueprint. The complete graph of nodes that defines how a request is processed.

- Versioned and publishable
- Has a unique, stable ID
- The unit you build, test, and deploy
- Can be re-used across ChatKit, SDK, or API

Node

The building block. Each node does exactly one thing.

- Typed inputs and outputs
- Connections enforce data contracts
- Executes independently — traceable
- Chain nodes to compose complex behavior

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A workflow is the graph. A node is the vertex. Edges carry typed data.

Node Types

Start

Entry point — accepts initial user input

Agent

Calls an LLM with instructions + tools

Classifier

Routes to one of N branches based on output category

If / Else

Conditional logic on data values

File Search

Vector search over uploaded documents (RAG)

Transform

Maps data between incompatible node outputs/inputs

MCP

Calls a hosted MCP server tool (third-party data/actions)

Guardrails

Input/output safety validation before/after agents

End

Returns final output to the caller

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Typed connections — each node declares its output schema; the canvas enforces that downstream nodes receive what they expect.

Templates vs Blank Canvas

Start from a Template

Best when: your use case fits a known pattern

- Support triage with routing
- Document Q&A (RAG)
- Research assistant
- Multi-step approval workflow

Pre-wired nodes — modify instructions, swap models, go.

Start Blank

Best when: designing something novel

- Custom multi-agent topology
- Unique data transformation pipeline
- Experimental classifier routing
- Prototyping from first principles

Add nodes one by one — full control over the graph.

Preview & Testing

Agent Builder has a built-in interactive test panel — no deploy step required.

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Interactive Runs

Send test inputs and see full output in real time



File Attachments

Attach PDFs and documents for File Search workflows



Per-Node Traces

Inspect what each node received and returned

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Attach PDFs and documents for File Search workflows



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Demo 1 (Course Assistant) — the simplest test case: Start → Agent → End. Per-node trace shows exactly what the agent saw and returned.

Guardrails — Open-Source Safety Layer

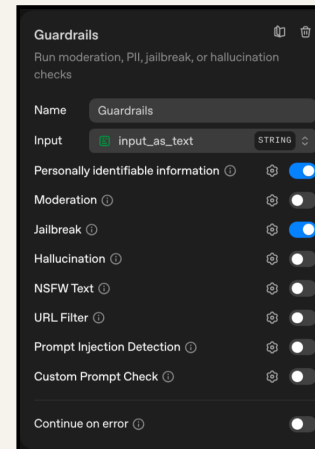
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What It Protects Against

- PII detection — Block or Redact mode
- Jailbreak attempts
- Prompt injection attacks
- Off-topic requests
- Sensitive data leakage in responses

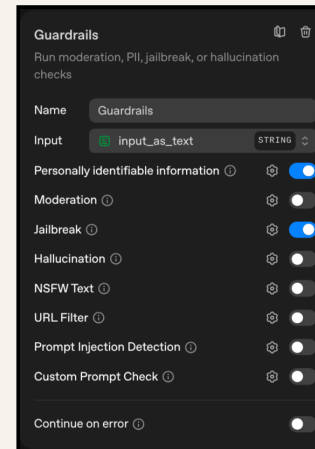


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Demo 2 (Support Triage) — Guardrails node gates every request before the Classifier. PII set to Block mode; jailbreak detection enabled.

MCP Integration

MCP nodes connect your workflow to hosted third-party tools via the Model Context Protocol.

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How It Works

- Add an MCP node to any workflow
- Pick a server from the Connector Registry
- The agent gains access to that tool set
- Examples: Slack, GitHub, Google Drive, custom APIs

▲ Context-Window Cost

Each MCP tool you add expands the model's context window with tool definitions.

Stack wisely: more tools = higher latency & cost per call.

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Demo 3 (File Search RAG) uses the built-in File Search node — same composition pattern as MCP, but backed by a vector store instead of an external API.

Widget Studio

Go beyond plain text — render **rich, structured outputs** directly from agent responses.

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Tables & Cards

Structured data rendered for review



Forms

Approval and input UIs from agent output



Custom Layouts

Enterprise dashboards, data review UIs

Widget Studio

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Enterprise dashboards, data review UIs

Widget Studio is configured in the workflow — the agent outputs structured JSON; Widget Studio renders it as a component in ChatKit.

Versioning & Publishing

Workflows are **first-class versioned objects** — not just scripts you save and overwrite.

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Each Version Has

- A unique, stable workflow ID
- A published deployment URL / API endpoint
- An immutable snapshot of the graph

Workflow Lifecycle

- Prior versions remain accessible
- Safe rollbacks with no downtime
- Staged promotion (dev → staging → prod)
- Point ChatKit at a specific version ID

Deployment Paths

Once your workflow is published, two paths to ship it:

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ChatKit (recommended default)

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- Embed a chat UI in minutes
- Handles auth, streaming, file upload
- No backend agent loop code needed
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SDK Code Export

- Export as Python or TypeScript
- Full server-side orchestration control
- Custom approval flows & state management
- Tool execution hooks & multi-runtime

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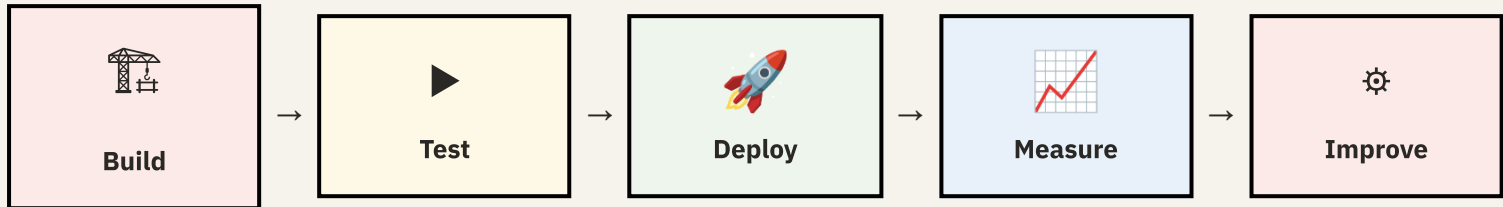
Start with ChatKit. Export to SDK only when you need capabilities the hosted path can't provide.

Evals Integration

Evals closes the loop between building and improving your workflow.

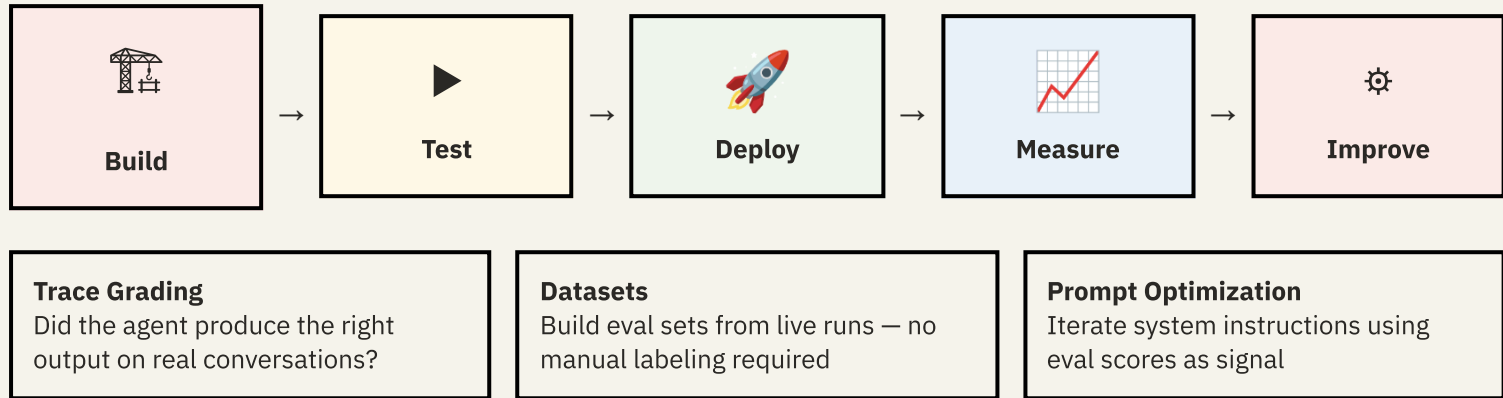
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Agent Builder vs Agents SDK

Reach for Agent Builder when...

- Visual builder for non-dev teammates
- Shipping to ChatKit is the goal
- A template is close to your use case
- You want Evals + iteration in the UI
- Prototyping and testing workflows fast

Reach for Agents SDK when...

- Server-side orchestration & custom state
- Custom human-in-the-loop approval flows
- Fine-grained tool execution control
- Multi-runtime or non-OpenAI deployment
- Full code ownership is a requirement

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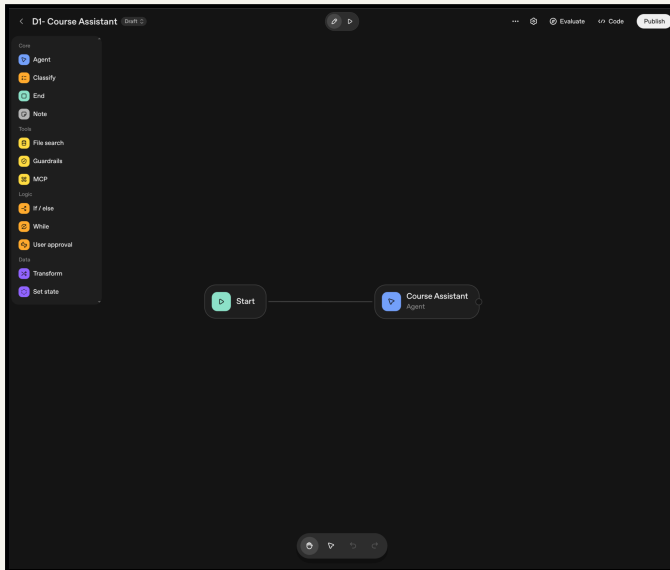
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These paths are **not mutually exclusive** — prototype in Agent Builder, export to SDK for production hardening.

Demo: Module 3 — Agent Builder Use Cases

Demo 1 — Course Assistant

Simplest workflow: Start → Agent → End



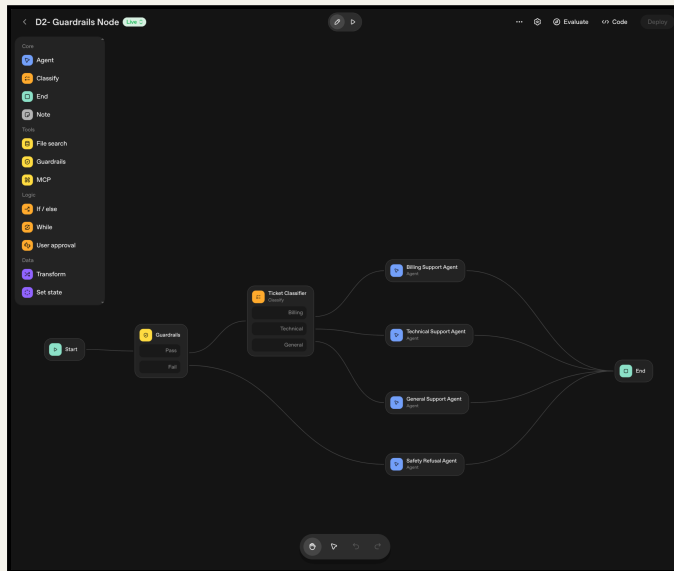
What It Demonstrates

- Minimal node composition — one Agent node
- How to set system instructions on a node
- Per-node trace inspection in preview mode
- Good starter template for any assistant

Model: gpt-5.4-mini with chat history enabled

Demo 2 — Support Triage with Guardrails

Multi-agent routing + safety: Guardrails → Classifier → 3 specialists (or Safety Refusal)

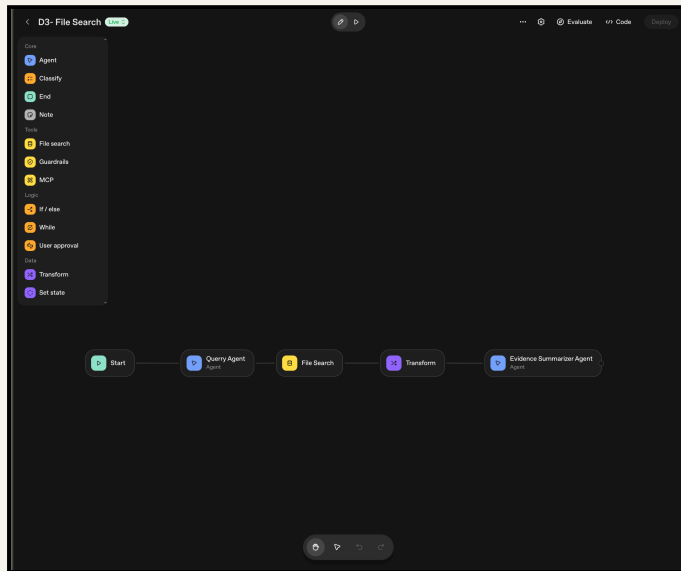


What It Demonstrates

- Guardrails node: PII block mode + jailbreak detection
- Classifier routing to Billing / Technical / General
- Safety Refusal Agent for policy violations
- Confidence-based routing prevents unsafe paths

Demo 3 — File Search RAG (SEC 10-K Filings)

Grounded answers from documents: Start → Query Agent → File Search → Transform → Evidence Summarizer



What It Demonstrates

- File Search node over 10 SEC 10-K filings
- Transform node: bridges typed output mismatch
- Evidence Summarizer with cited sources
- Vector store with real financial documents

Query: "What does NVIDIA say about AI data centers and supply chain risks?"

ChatKit — Embeddable Chat UI

Frontend-only — React components + framework-agnostic Web Component.

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```
import { ChatKit, useChatKit } from '@openai/chatkit-react';

const chatkit = useChatKit({
  api: { getClientSecret },
  .....
});

return <ChatKit control={chatkit.control} />;
```

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Includes: file upload, light/dark theming, tool invocation display, chain-of-thought visualization, suggestion chips, event system, and more.

[1][ChatKit — OpenAI Docs](#)

[2][chatkit-js \(GitHub\)](#)

ChatKit — Backend Token Exchange

```
// POST /api/create-session (your backend)
const response = await fetch(
  "https://api.openai.com/v1/chatkit/sessions",
  {
    method: "POST",
    headers: {
      ....
    }
  });

const { client_secret } = await response.json();
// Return client_secret to the frontend
```

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const { client_secret } = await response.json();
// Return client_secret to the frontend
```

Key pattern: Your server-side API key is exchanged for a short-lived client token. Per-user sessions via cookies. The frontend *never* sees your API key.

Demo: ChatKit App

When to Use Each Layer

Direct API Control

- **Responses API** — direct calls, full control, learning

Multi-Agent Systems

- **Agents SDK** — orchestration, custom logic, guardrails

Ship a Chat UI

- **ChatKit** — embed a chat experience fast

Visual Workflows & Evals

- **Agent Builder** — drag-and-drop, non-dev teams
- **Evals** — measure and improve agent quality

Key Takeaways

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- **Start with Responses API** for maximum control and the deepest learning
- **Add Agents SDK** when you need multi-agent orchestration, guardrails, tracing
- **Use AgentKit / ChatKit** when you're ready to ship to real users
- Each layer **builds on the previous** — no vendor lock-in at the SDK level
- The whole stack is available at **standard API pricing** — no platform fee

What We Built

From notebooks to full-stack agent apps across three modules:

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From notebooks to full-stack agent apps across three modules:

- **Paper Chat App** — interactive Q&A on uploaded PDFs (Module 1)
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- **Research Report App** — structured analysis exported as Markdown (Module 1)

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- **Video Script Generator** — paper → 60-second video script with DALL-E scenes (Module 1)
- **Research Report App** — structured analysis exported as Markdown (Module 1)
- **3 Agent Builder Workflows** — Course Assistant, Support Triage, File Search RAG (Module 3)

What We Built

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- **Research Report App** — structured analysis exported as Markdown (Module 1)
- **3 Agent Builder Workflows** — Course Assistant, Support Triage, File Search RAG (Module 3)

The patterns you learn here apply at every layer of the stack.

Q&A & References

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- [Responses API Reference](#) — OpenAI Docs
- [openai-agents-python](#) — GitHub
- [Agents SDK Documentation](#) — OpenAI
- [Introducing AgentKit](#) — OpenAI
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Questions before we move on?

